Chapter Contents

Chapter 14 - Second-Order Applications

The preceding chapters developed methods for solving second-order linear equations and selecting solutions from initial data. This chapter turns those methods into models of physical systems.

The main mathematical form is:

$$Mu'' + Cu' + Ku = F(t).$$

The interpretation of the symbols changes with the application:

- for a spring, u is displacement
- for a circuit, u may be charge
- for a floating body, u is vertical displacement from equilibrium

The equation should not be memorized as a string of symbols. Each term must be connected to a physical law, a sign convention, and a unit.

The main questions are:

- how does Newton's second law produce the spring equation?
- why does gravity disappear when displacement is measured from equilibrium?
- what distinguishes undamped, underdamped, critically damped, and overdamped motion?
- how are circular frequency, ordinary frequency, period, amplitude, and phase related?
- what is the difference between transient and steady-state response?
- why does matching a natural frequency create resonance in an ideal undamped system?
- how does an RLC circuit produce the same differential-equation structure?
- how does Archimedes' principle create a buoyant restoring force?

All examples, equations, and exercises in this chapter are independently constructed. The reference text is used only to identify the mathematical topics.

From Physical Laws To A Differential Equation

The Goal Of This Block

The goal is to construct the mass-spring equation from forces rather than quote it. Every sign will be tied to one declared positive direction.

Choose The Coordinate Before Writing Forces

Let $x(t)$ be the displacement of a mass from its **equilibrium position**. Choose downward displacement as positive.

Then:

- $x > 0$ means the mass is below equilibrium
- $x < 0$ means the mass is above equilibrium
- $x' > 0$ means the mass is moving downward
- $x' < 0$ means the mass is moving upward

This choice is not the only valid convention. Upward could be positive instead. What matters is using one convention consistently in the force equation and the initial data.

The Spring Restoring Force

Hooke's law says that the magnitude of the spring force is proportional to displacement from equilibrium:

$$|F_s| = k|x|,$$

where $k > 0$ is the spring constant.

The spring force points back toward equilibrium. With downward positive:

- if $x > 0$, the force points upward and is negative
- if $x < 0$, the force points downward and is positive

Both cases are represented by:

$$\boxed{F_s = -kx}.$$

The minus sign expresses direction, not a negative spring constant.

The Damping Force

For linear viscous damping, the resistive force is proportional to velocity and points opposite the motion.

If $c \geq 0$ is the damping coefficient, then:

$$\boxed{F_d = -cx'}.$$

When $x' > 0$, the mass moves downward and the damping force is upward. When $x' < 0$, the mass moves upward and the damping force is downward. The same formula handles both cases.

Newton's Second Law

Let $F(t)$ be an externally applied force, measured positive downward. The three forces measured from equilibrium are:

$$F_s = -kx, \qquad F_d = -cx', \qquad F_{\text{ext}} = F(t).$$

Newton's second law is:

$$\text{mass} \times \text{acceleration} = \text{sum of forces}.$$

Therefore:

$$mx'' = -kx - cx' + F(t).$$

Add cx' and kx to both sides:

$$\boxed{mx'' + cx' + kx = F(t)}.$$

Each term has the units of force:

- mx'' is the inertial term
- cx' is the damping term
- kx is the restoring term
- $F(t)$ is the applied forcing

Where Did Gravity Go?

Gravity has not been ignored. It is cancelled by the spring's static extension when displacement is measured from equilibrium.

Let $z(t)$ be the total downward extension from the spring's natural length. Ignoring damping briefly, Newton's law is:

$$mz'' = mg - kz + F(t).$$

Let ℓ be the static extension produced by the mass. At equilibrium, acceleration is zero and no external force is present, so:

$$mg - k\ell = 0.$$

Therefore:

$$\boxed{k\ell = mg}.$$

Now measure displacement x from equilibrium:

$$z = \ell + x.$$

Because ℓ is constant:

$$z' = x', \qquad z'' = x''.$$

Substitute $z = \ell + x$ into the natural-length equation:

$$mx'' = mg - k(\ell + x) + F(t).$$

Expand the spring term:

$$mx'' = mg - k\ell - kx + F(t).$$

Use $k\ell = mg$:

$$mx'' = mg - mg - kx + F(t).$$

The gravity and static-spring terms cancel:

$$mx'' = -kx + F(t).$$

Including damping restores the full model:

$$mx'' + cx' + kx = F(t).$$

Mass, Weight, And Units

In SI units:

- mass is measured in kilograms
- force is measured in newtons
- displacement is measured in metres
- k is measured in newtons per metre
- c is measured in newton-seconds per metre

If a problem gives a mass m and a static extension ℓ, then:

$$k = \frac{mg}{\ell}.$$

Weight is the force mg, not the mass m.

In imperial units, a weight in pounds-force must be converted to mass in slugs:

$$m = \frac{\text{weight}}{g}.$$

Mixing weight and mass produces a dimensionally incorrect inertial term.

A Small Static-Stretch Example

A 3 kg mass stretches a vertical spring by 0.147 m. Using $g = 9.8\ \mathrm{m/s^2}$, find k.

At static equilibrium:

$$k\ell = mg.$$

Solve for k:

$$k = \frac{mg}{\ell}.$$

Substitute $m = 3$, $g = 9.8$, and $\ell = 0.147$:

$$\begin{aligned} k &= \frac{3(9.8)}{0.147} \\ &= \frac{29.4}{0.147} \\ &= 200. \end{aligned}$$

Therefore:

$$\boxed{k = 200\ \mathrm{N/m}}.$$

Block 1 Summary

With displacement measured from equilibrium and downward chosen positive:

$$F_s = -kx, \qquad F_d = -cx'.$$

Newton's second law gives:

$$\boxed{mx'' + cx' + kx = F(t)}.$$

Gravity is absent from this equilibrium-coordinate equation because the static balance $k\ell = mg$ has already accounted for it.

Free Undamped Motion And Harmonic Quantities

The Goal Of This Block

The goal is to solve the free undamped spring and interpret every constant in the resulting harmonic motion.

The Free Undamped Equation

Free motion means no external force:

$$F(t) = 0.$$

Undamped motion means:

$$c = 0.$$

The spring equation becomes:

$$mx'' + kx = 0.$$

Divide by $m > 0$:

$$x'' + \frac{k}{m}x = 0.$$

Define the natural circular frequency:

$$\boxed{\omega_0 = \sqrt{\frac{k}{m}}}.$$

Then:

$$x'' + \omega_0^2 x = 0.$$

The general solution is:

$$\boxed{x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)}.$$

Circular Frequency, Frequency, And Period

The circular frequency ω_0 is measured in radians per unit time. The ordinary frequency f counts cycles per unit time:

$$\boxed{f = \frac{\omega_0}{2\pi}}.$$

The period T is the time for one complete cycle:

$$\boxed{T = \frac{1}{f} = \frac{2\pi}{\omega_0}}.$$

These are different quantities. A circular frequency of 5 rad/s is not a frequency of 5 cycles/s.

Amplitude And Phase

The combination:

$$A\cos(\omega t) + B\sin(\omega t)$$

can be written as one shifted cosine:

$$\boxed{R\cos(\omega t-\delta)}.$$

Expand the shifted cosine:

$$R\cos(\omega t-\delta)=R\cos(\omega t)\cos\delta+R\sin(\omega t)\sin\delta.$$

Compare coefficients with $A\cos(\omega t)+B\sin(\omega t)$:

$$A=R\cos\delta,\qquad B=R\sin\delta.$$

Square and add:

$$A^2+B^2=R^2(\cos^2\delta+\sin^2\delta).$$

Since $\cos^2\delta+\sin^2\delta=1$:

$$\boxed{R=\sqrt{A^2+B^2}}.$$

The phase should be chosen from both coefficient signs:

$$\cos\delta=\frac{A}{R},\qquad \sin\delta=\frac{B}{R}.$$

Using a two-argument inverse tangent, often written $\operatorname{atan2}(B,A)$, places δ in the correct quadrant.

A Complete Free Undamped Example

A 2 kg mass is attached to a spring with $k = 50$ N/m. There is no damping or external force. Downward is positive. The mass begins 0.08 m below equilibrium with an upward velocity of 0.30 m/s.

Find the motion, amplitude, circular frequency, ordinary frequency, and period.

Step 1: Translate the physical data. The parameters are:

$$m = 2, \qquad k = 50, \qquad c = 0, \qquad F(t) = 0.$$

The initial displacement is downward and therefore positive:

$$x(0) = 0.08.$$

The initial velocity is upward and therefore negative:

$$x'(0) = -0.30.$$

Step 2: Construct and normalize the equation. Use $mx'' + kx = 0$:

$$2x'' + 50x = 0.$$

Divide by 2:

$$x'' + 25x = 0.$$

Thus:

$$\omega_0^2 = 25, \qquad \omega_0 = 5 \text{ rad/s}.$$

Step 3: Write the general motion. The solution of $x'' + 25x = 0$ is:

$$x = A\cos(5t) + B\sin(5t).$$

Differentiate:

$$x' = -5A\sin(5t) + 5B\cos(5t).$$

Step 4: Apply the displacement condition. Substitute $t = 0$ into the motion:

$$0.08 = A\cos(0) + B\sin(0).$$

Therefore:

$$A = 0.08.$$

Step 5: Apply the velocity condition. Substitute $t = 0$ into the velocity:

$$-0.30 = -5A\sin(0) + 5B\cos(0).$$

Thus:

$$-0.30 = 5B.$$

Divide by 5:

$$B = -0.06.$$

The motion is:

$$\boxed{x(t) = 0.08\cos(5t) - 0.06\sin(5t)}.$$

Step 6: Find the amplitude. Use $R = \sqrt{A^2 + B^2}$:

$$\begin{aligned} R &= \sqrt{(0.08)^2 + (-0.06)^2} \\ &= \sqrt{0.0064 + 0.0036} \\ &= \sqrt{0.01} \\ &= 0.10. \end{aligned}$$

Therefore the amplitude is:

$$\boxed{R = 0.10 \text{ m}}.$$

For the phase:

$$\cos\delta = \frac{0.08}{0.10} = \frac{4}{5}, \qquad \sin\delta = \frac{-0.06}{0.10} = -\frac{3}{5}.$$

One suitable phase is:

$$\delta = -\arctan\left(\frac{3}{4}\right).$$

Hence an equivalent form is:

$$\boxed{x(t) = 0.10\cos\left(5t + \arctan\frac{3}{4}\right)}.$$

Step 7: Find frequency and period. The ordinary frequency is:

$$\boxed{f = \frac{5}{2\pi}\ \mathrm{Hz}}.$$

The period is:

$$\boxed{T = \frac{2\pi}{5}\ \mathrm{s}}.$$

Block 2 Summary

For free undamped motion:

$$x'' + \omega_0^2 x = 0, \qquad \omega_0 = \sqrt{\frac{k}{m}}.$$

The motion is harmonic:

$$x = A\cos(\omega_0 t) + B\sin(\omega_0 t).$$

Its amplitude, frequency, and period are:

$$R = \sqrt{A^2 + B^2}, \qquad f = \frac{\omega_0}{2\pi}, \qquad T = \frac{2\pi}{\omega_0}.$$

Damping And The Three Free-Motion Regimes

The Goal Of This Block

The goal is to classify free damped motion directly from the characteristic discriminant and connect each root pattern to its physical behavior.

The Free Damped Equation

For free motion, $F(t) = 0$. With damping $c > 0$:

$$mx'' + cx' + kx = 0.$$

The characteristic equation is:

$$mr^2 + cr + k = 0.$$

The quadratic formula gives:

$$r = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

Define the discriminant:

$$\boxed{\Delta = c^2 - 4mk}.$$

The sign of Δ determines the type of free damped motion.

Overdamped Motion

If:

$$\Delta > 0,$$

the roots are real and distinct. The motion has the form:

$$x = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

Both roots are negative when $m, c, k > 0$. The motion returns toward equilibrium without sustained oscillation.

The physical interpretation is that resistance is stronger than the critical amount needed to suppress oscillation.

Critically Damped Motion

If:

$$\Delta = 0,$$

the repeated root is:

$$r = -\frac{c}{2m}.$$

The motion is:

$$x = (c_1 + c_2 t)e^{-ct/(2m)}.$$

Critical damping is the boundary between oscillatory and nonoscillatory return. It produces the fastest idealized return to equilibrium without oscillation.

The critical damping coefficient is found from $c^2 = 4mk$:

$$\boxed{c_{\text{crit}} = 2\sqrt{mk}}.$$

Underdamped Motion

If:

$$\Delta < 0,$$

the roots are complex:

$$r = -\frac{c}{2m} \pm i\omega_d,$$

where the damped circular frequency is:

$$\boxed{\omega_d = \frac{\sqrt{4mk - c^2}}{2m}}.$$

The motion is:

$$x = e^{-ct/(2m)}[c_1 \cos(\omega_d t) + c_2 \sin(\omega_d t)].$$

The sine and cosine produce oscillation. The exponential envelope shrinks the amplitude toward zero.

A Three-Case Comparison

Let:

$$m = 1, \qquad k = 9.$$

Then:

$$c_{\text{crit}} = 2\sqrt{(1)(9)} = 6.$$

Case 1: $c = 2$. The discriminant is:

$$\Delta = 2^2 - 4(1)(9) = 4 - 36 = -32 < 0.$$

The system is underdamped.

Case 2: $c = 6$. The discriminant is:

$$\Delta = 6^2 - 36 = 0.$$

The system is critically damped.

Case 3: $c = 8$. The discriminant is:

$$\Delta = 8^2 - 36 = 28 > 0.$$

The system is overdamped.

The mass and spring are unchanged. Only the damping coefficient changes the qualitative motion.

A Complete Underdamped Example

Solve:

$$x'' + 2x' + 9x = 0,$$

subject to:

$$x(0) = 1, \qquad x'(0) = 0.$$

Step 1: Classify the roots. Here $m = 1, c = 2, k = 9$. The discriminant is:

$$\Delta = 2^2 - 4(1)(9) = -32 < 0.$$

Therefore the motion is underdamped.

The characteristic roots are:

$$\begin{aligned} r &= \frac{-2 \pm \sqrt{-32}}{2} \\ &= -1 \pm 2\sqrt{2}\, i. \end{aligned}$$

Step 2: Write the general motion. Let:

$$\beta = 2\sqrt{2}.$$

Then:

$$x = e^{-t}[c_1 \cos(\beta t) + c_2 \sin(\beta t)].$$

Step 3: Apply $x(0) = 1$. Substitute $t = 0$:

$$1 = e^0[c_1 \cos 0 + c_2 \sin 0].$$

Thus:

$$c_1 = 1.$$

Step 4: Differentiate before applying the velocity. Write:

$$f(t) = c_1 \cos(\beta t) + c_2 \sin(\beta t).$$

Then:

$$x = e^{-t} f(t).$$

Differentiate with the product rule:

$$x' = e^{-t}[f'(t) - f(t)].$$

Also:

$$f'(t) = -\beta c_1 \sin(\beta t) + \beta c_2 \cos(\beta t).$$

At $t = 0$:

$$f(0) = c_1, \qquad f'(0) = \beta c_2.$$

Therefore:

$$x'(0) = \beta c_2 - c_1.$$

Apply $x'(0) = 0$ and $c_1 = 1$:

$$0 = \beta c_2 - 1.$$

Add 1:

$$\beta c_2 = 1.$$

Divide by $\beta = 2\sqrt{2}$:

$$c_2 = \frac{1}{2\sqrt{2}}.$$

Step 5: Write and interpret the motion.

$$\boxed{x(t) = e^{-t}\left[\cos(2\sqrt{2}\,t) + \frac{1}{2\sqrt{2}}\sin(2\sqrt{2}\,t)\right]}.$$

The oscillatory bracket remains bounded, while $e^{-t} \to 0$. Therefore:

$$x(t) \to 0 \qquad \text{as} \qquad t \to \infty.$$

The free damped response is transient.

Block 3 Summary

For free damped motion:

$$mx'' + cx' + kx = 0,$$

the discriminant:

$$\Delta = c^2 - 4mk$$

classifies the response:

- $\Delta > 0$: overdamped
- $\Delta = 0$: critically damped
- $\Delta < 0$: underdamped

For positive m, c, k, every free damped solution tends to equilibrium.

Forced Motion, Steady State, And Resonance

The Goal Of This Block

The goal is to separate a forced response into transient and steady-state parts and understand why ideal resonance produces an unbounded amplitude.

Free Versus Forced Response

For:

$$mx'' + cx' + kx = F(t),$$

the complete solution is:

$$x = x_h + x_p.$$

The homogeneous part x_h is the system's free response. For a damped system, it normally contains decaying exponentials and is called the **transient** part.

The particular part x_p is generated by the continuing external input. For a bounded sinusoidal input in a damped system, it normally becomes the **steady-state** part.

The names describe long-term behavior:

$$x_{\mathrm{tr}}(t) \to 0, \qquad x_{\mathrm{ss}}(t) \not\to 0.$$

A Complete Damped Forced Example

Solve:

$$x'' + 2x' + 5x = 10\cos(2t),$$

subject to:

$$x(0) = 0, \qquad x'(0) = 0.$$

Step 1: Solve the homogeneous equation. The characteristic equation is:

$$r^2 + 2r + 5 = 0.$$

The roots are:

$$r = -1 \pm 2i.$$

Therefore:

$$x_h = e^{-t}[c_1\cos(2t) + c_2\sin(2t)].$$

Step 2: Build the sinusoidal trial. Use:

$$x_p = A\cos(2t) + B\sin(2t).$$

Differentiate:

$$x_p' = -2A\sin(2t) + 2B\cos(2t).$$

Differentiate again:

$$x_p'' = -4A\cos(2t) - 4B\sin(2t).$$

Step 3: Substitute and group cosine terms. Substitute into $x_p'' + 2x_p' + 5x_p$:

$$\begin{aligned}&[-4A\cos(2t) - 4B\sin(2t)]\\ &\quad + 2[-2A\sin(2t) + 2B\cos(2t)]\\ &\quad + 5[A\cos(2t) + B\sin(2t)].\end{aligned}$$

The cosine coefficient is:

$$-4A + 4B + 5A = A + 4B.$$

The sine coefficient is:

$$-4B - 4A + 5B = B - 4A.$$

Therefore:

$$x_p'' + 2x_p' + 5x_p = (A + 4B)\cos(2t) + (B - 4A)\sin(2t).$$

Step 4: Match the forcing coefficients. The required forcing is:

$$10\cos(2t) + 0\sin(2t).$$

Hence:

$$\begin{aligned}A + 4B &= 10,\\ B - 4A &= 0.\end{aligned}$$

The second equation gives:

$$B = 4A.$$

Substitute into the first equation:

$$A + 4(4A) = 10.$$

Simplify:

$$17A = 10.$$

Therefore:

$$A = \frac{10}{17}.$$

Then:

$$B = 4A = \frac{40}{17}.$$

Thus:

$$x_p = \frac{10}{17}\cos(2t) + \frac{40}{17}\sin(2t).$$

Step 5: Write the complete solution.

$$x = e^{-t}[c_1\cos(2t) + c_2\sin(2t)] + \frac{10}{17}\cos(2t) + \frac{40}{17}\sin(2t).$$

Step 6: Apply $x(0) = 0$. At $t = 0$:

$$0 = c_1 + \frac{10}{17}.$$

Therefore:

$$c_1 = -\frac{10}{17}.$$

Step 7: Differentiate locally and apply $x'(0) = 0$. Let:

$$f(t) = c_1 \cos(2t) + c_2 \sin(2t).$$

Then:

$$x_h = e^{-t} f(t),$$

so:

$$x_h' = e^{-t}[f'(t) - f(t)].$$

At $t = 0$:

$$f(0) = c_1, \qquad f'(0) = 2c_2.$$

Therefore:

$$x_h'(0) = 2c_2 - c_1.$$

For the particular solution:

$$x_p' = -\frac{20}{17}\sin(2t) + \frac{80}{17}\cos(2t).$$

Thus:

$$x_p'(0) = \frac{80}{17}.$$

Apply the complete velocity condition:

$$0 = 2c_2 - c_1 + \frac{80}{17}.$$

Substitute $c_1 = -10/17$:

$$0 = 2c_2 + \frac{10}{17} + \frac{80}{17}.$$

Combine the fractions:

$$0 = 2c_2 + \frac{90}{17}.$$

Subtract $90/17$:

$$2c_2 = -\frac{90}{17}.$$

Divide by 2:

$$c_2 = -\frac{45}{17}.$$

Step 8: Identify transient and steady-state parts. The selected motion is:

$$\boxed{x(t) = -\frac{e^{-t}}{17}[10\cos(2t) + 45\sin(2t)] + \frac{10}{17}\cos(2t) + \frac{40}{17}\sin(2t)}.$$

The transient part is:

$$x_{\text{tr}} = -\frac{e^{-t}}{17}[10\cos(2t) + 45\sin(2t)].$$

It tends to zero because of e^{-t}.

The steady-state part is:

$$x_{\text{ss}} = \frac{10}{17}\cos(2t) + \frac{40}{17}\sin(2t).$$

Steady-State Amplitude And Phase Lag

For the previous steady-state response:

$$A = \frac{10}{17}, \qquad B = \frac{40}{17}.$$

The amplitude is:

$$\begin{aligned} R &= \sqrt{A^2 + B^2} \\ &= \sqrt{\frac{100 + 1600}{289}} \\ &= \frac{10\sqrt{17}}{17} \\ &= \frac{10}{\sqrt{17}}. \end{aligned}$$

Because both coefficients are positive, choose:

$$\delta = \arctan\left(\frac{B}{A}\right) = \arctan(4).$$

Therefore:

$$\boxed{x_{\text{ss}}(t) = \frac{10}{\sqrt{17}}\cos[2t - \arctan(4)]}.$$

The response oscillates at the forcing frequency, but its peak occurs later than the peak of $10\cos(2t)$. This shift is the phase lag.

Pure Resonance In An Undamped System

Consider the zero-data IVP:

$$x'' + 9x = 6\cos(3t), \qquad x(0) = 0, \qquad x'(0) = 0.$$

The natural circular frequency is:

$$\omega_0 = 3.$$

The forcing also has circular frequency 3. The usual trial $A\cos(3t) + B\sin(3t)$ overlaps the homogeneous solution, so use a resonant trial.

The operator identity needed here is:

$$\left(\frac{d^2}{dt^2} + 9\right)[t\sin(3t)] = 6\cos(3t).$$

Therefore one particular solution is:

$$x_p = t\sin(3t).$$

It also satisfies the zero initial data:

$$x_p(0) = 0.$$

Differentiate:

$$x_p' = \sin(3t) + 3t\cos(3t).$$

Thus:

$$x_p'(0) = 0.$$

The selected solution is therefore:

$$\boxed{x(t) = t\sin(3t)}.$$

The factor t makes the oscillation envelope grow without bound. This is **pure resonance** in the ideal undamped model.

Damping changes the model and prevents this unlimited linear growth for a bounded sinusoidal input.

Block 4 Summary

For a damped forced system:

$$x = x_{\text{tr}} + x_{\text{ss}}.$$

The transient part comes from the homogeneous equation and decays. The steady-state part follows the continuing forcing.

In an ideal undamped system, forcing at the natural circular frequency creates a resonant term with a growing factor such as t.

Series RLC Circuits

The Goal Of This Block

The goal is to derive the charge equation for a series resistor-inductor-capacitor circuit and connect its terms to the mass-spring model.

Charge And Current

Let:

- $q(t)$ be charge on the capacitor, measured in coulombs
- $I(t)$ be current, measured in amperes

Current is the time derivative of charge:

$$\boxed{I = q'}.$$

Therefore:

$$I' = q''.$$

The distinction matters. An initial charge condition specifies $q(0)$, while an initial current condition specifies $q'(0)$.

Voltage Drops Around The Loop

For a series RLC circuit:

- the inductor voltage drop is LI'
- the resistor voltage drop is RI
- the capacitor voltage drop is q/C
- the applied voltage is $E(t)$

Kirchhoff's voltage law gives:

$$LI' + RI + \frac{q}{C} = E(t).$$

Substitute $I = q'$ and $I' = q''$:

$$\boxed{Lq'' + Rq' + \frac{1}{C}q = E(t)}.$$

Here:

- L is inductance in henries
- R is resistance in ohms
- C is capacitance in farads
- $E(t)$ is applied voltage in volts

The Mechanical-Electrical Analogy

Compare:

$$mx'' + cx' + kx = F(t)$$

with:

$$Lq'' + Rq' + \frac{1}{C}q = E(t).$$

The corresponding quantities are:

mechanical system	electrical system
m	L
c	R
k	$1/C$
x	q
x'	I
$F(t)$	$E(t)$

The analogy is structural. Mass and inductance do not have the same physical units, but they occupy the same mathematical position in their equations.

A Complete Charge-And-Current Example

A series circuit has:

$$L = 1\,\mathrm{H}, \qquad R = 4\,\Omega, \qquad C = \frac{1}{5}\,\mathrm{F},$$

and a constant applied voltage:

$$E(t) = 10\,\mathrm{V}.$$

Initially:

$$q(0) = 0, \qquad I(0) = 0.$$

Find the charge and current.

Step 1: Construct the charge equation. Use:

$$Lq'' + Rq' + \frac{1}{C}q = E(t).$$

Since $1/C = 5$, substitution gives:

$$q'' + 4q' + 5q = 10.$$

The current condition becomes:

$$q'(0) = I(0) = 0.$$

Step 2: Solve the homogeneous equation. The characteristic equation is:

$$r^2 + 4r + 5 = 0.$$

The roots are:

$$r = -2 \pm i.$$

Therefore:

$$q_h = e^{-2t}[c_1 \cos t + c_2 \sin t].$$

Step 3: Find a constant particular charge. Try:

$$q_p = A.$$

Then:

$$q_p' = 0, \qquad q_p'' = 0.$$

Substitute into $q'' + 4q' + 5q = 10$:

$$5A = 10.$$

Divide by 5:

$$A = 2.$$

Thus:

$$q_p = 2.$$

Step 4: Write the complete charge.

$$q = e^{-2t}[c_1 \cos t + c_2 \sin t] + 2.$$

Step 5: Apply $q(0) = 0$. Substitute $t = 0$:

$$0 = c_1 + 2.$$

Therefore:

$$c_1 = -2.$$

Step 6: Differentiate to obtain current. Let:

$$f(t) = c_1 \cos t + c_2 \sin t.$$

Then:

$$q = e^{-2t} f(t) + 2.$$

Differentiate:

$$q' = e^{-2t}[f'(t) - 2f(t)].$$

At $t = 0$:

$$f(0) = c_1, \qquad f'(0) = c_2.$$

Thus:

$$q'(0) = c_2 - 2c_1.$$

Apply $q'(0) = 0$ and $c_1 = -2$:

$$0 = c_2 - 2(-2).$$

Therefore:

$$c_2 = -4.$$

The charge is:

$$\boxed{q(t) = 2 - e^{-2t}[2\cos t + 4\sin t]}.$$

Step 7: Simplify the current. Differentiate the selected charge. Let:

$$f(t) = -2\cos t - 4\sin t.$$

Then:

$$q = 2 + e^{-2t} f(t).$$

Calculate:

$$f'(t) = 2\sin t - 4\cos t.$$

Use $q' = e^{-2t}[f' - 2f]$:

$$\begin{aligned} I(t) = q'(t) &= e^{-2t}[2\sin t - 4\cos t + 4\cos t + 8\sin t] \\ &= 10e^{-2t}\sin t. \end{aligned}$$

Therefore:

$$\boxed{I(t) = 10e^{-2t}\sin t}.$$

As $t \to \infty$:

$$q(t) \to 2, \qquad I(t) \to 0.$$

The capacitor approaches a steady charge while the current dies away.

Writing An Equation Directly For Current

Differentiate the charge equation:

$$Lq'' + Rq' + \frac{1}{C}q = E(t).$$

Using $I = q'$ gives:

$$\boxed{LI'' + RI' + \frac{1}{C}I = E'(t)}.$$

An initial value for I' is not usually given directly. It must be recovered from the original loop equation:

$$LI'(0) + RI(0) + \frac{q(0)}{C} = E(0).$$

Solve for $I'(0)$:

$$\boxed{I'(0) = \frac{E(0) - RI(0) - q(0)/C}{L}}.$$

Solving for charge first is often simpler because $q(0)$ and $I(0) = q'(0)$ are already the natural initial conditions.

Block 5 Summary

For a series RLC circuit:

$$\boxed{Lq'' + Rq' + \frac{1}{C}q = E(t)}, \qquad I = q'.$$

Inductance corresponds mathematically to mass, resistance to damping, inverse capacitance to spring stiffness, and applied voltage to external force.

Buoyancy And Floating-Body Oscillations

The Goal Of This Block

The goal is to derive a restoring force from Archimedes' principle and show why a slightly displaced floating body behaves like an undamped spring.

Equilibrium Of A Vertical Cylinder

Consider a vertical cylinder with:

- mass m
- constant horizontal cross-sectional area A
- submerged equilibrium depth h
- liquid mass density ρ

The cylinder's weight is:

$$mg.$$

At equilibrium, the displaced liquid volume is:

$$Ah.$$

The mass of displaced liquid is ρAh, so the upward buoyant force is:

$$\rho gAh.$$

Equilibrium requires upward buoyancy to equal downward weight:

$$\rho gAh = mg.$$

Cancel $g > 0$:

$$\rho A h = m.$$

Solve for the submerged depth:

$$\boxed{h = \frac{m}{\rho A}}.$$

The Buoyant Restoring Force

Measure displacement $x(t)$ upward from equilibrium and choose upward as positive.

If the cylinder moves upward by $x > 0$, its submerged depth decreases from h to:

$$h - x.$$

The new buoyant force is:

$$F_b = \rho g A(h - x).$$

The weight remains downward:

$$F_g = -mg.$$

Newton's second law gives:

$$mx'' = \rho g A(h - x) - mg.$$

Expand:

$$mx'' = \rho g A h - \rho g A x - mg.$$

Use the equilibrium identity $\rho g A h = mg$:

$$mx'' = mg - \rho g A x - mg.$$

The equilibrium terms cancel:

$$mx'' = -\rho g A x.$$

Move the restoring term to the left:

$$\boxed{mx'' + \rho g A x = 0}.$$

This has the same form as an undamped spring with effective stiffness:

$$\boxed{k_\mathrm{b} = \rho g A}.$$

Natural Frequency And Period

Divide the buoyancy equation by m:

$$x'' + \frac{\rho g A}{m} x = 0.$$

Therefore the natural circular frequency is:

$$\boxed{\omega_0 = \sqrt{\frac{\rho g A}{m}}}.$$

The period is:

$$\boxed{T = 2\pi\sqrt{\frac{m}{\rho g A}}}.$$

This model assumes the cylinder remains vertical, the cross-sectional area at the waterline stays constant, and the displacement is small enough that the body remains partially submerged.

A Complete Buoyancy Example

A vertical cylinder has:

$$m = 9.8\ \mathrm{kg}, \qquad A = 0.05\ \mathrm{m}^2.$$

It floats in water with:

$$\rho = 1000\ \mathrm{kg/m^3}, \qquad g = 9.8\ \mathrm{m/s^2}.$$

It is raised 0.04 m above equilibrium and released from rest. Find the equilibrium submerged depth and subsequent motion.

Step 1: Find the equilibrium submerged depth. Use:

$$h = \frac{m}{\rho A}.$$

Substitute:

$$\begin{aligned} h &= \frac{9.8}{1000(0.05)} \\ &= \frac{9.8}{50} \\ &= 0.196. \end{aligned}$$

Therefore:

$$\boxed{h = 0.196 \text{ m}}.$$

Step 2: Construct the motion equation. Use:

$$mx'' + \rho g A x = 0.$$

Substitute the parameters:

$$9.8x'' + 1000(9.8)(0.05)x = 0.$$

Calculate the restoring coefficient:

$$1000(9.8)(0.05) = 490.$$

Thus:

$$9.8x'' + 490x = 0.$$

Divide by 9.8:

$$x'' + 50x = 0.$$

The natural circular frequency is:

$$\omega_0 = \sqrt{50} = 5\sqrt{2}.$$

Step 3: Translate the initial data. Upward is positive. The cylinder is raised by 0.04 m, so:

$$x(0) = 0.04.$$

It is released from rest, so:

$$x'(0) = 0.$$

Step 4: Solve the IVP. The general motion is:

$$x = c_1 \cos(5\sqrt{2}\,t) + c_2 \sin(5\sqrt{2}\,t).$$

Apply $x(0) = 0.04$:

$$c_1 = 0.04.$$

Differentiate:

$$x' = -5\sqrt{2}\,c_1 \sin(5\sqrt{2}\,t) + 5\sqrt{2}\,c_2 \cos(5\sqrt{2}\,t).$$

Apply $x'(0) = 0$:

$$0 = 5\sqrt{2}\,c_2.$$

Therefore:

$$c_2 = 0.$$

The motion is:

$$\boxed{x(t) = 0.04\cos(5\sqrt{2}\,t)}.$$

The period is:

$$\boxed{T = \frac{2\pi}{5\sqrt{2}}\ \mathrm{s}}.$$

Block 6 Summary

For a vertical cylinder of constant cross-sectional area A:

$$h = \frac{m}{\rho A}$$

and small vertical displacement from equilibrium satisfies:

$$\boxed{mx'' + \rho g A x = 0}.$$

The buoyant restoring force is proportional to the change in displaced volume.

A Unified Workflow And Common Mistakes

The Shared Second-Order Structure

Springs, circuits, and buoyancy models all fit:

$$\boxed{Mu'' + Cu' + Ku = F(t)}.$$

The roles are:

	spring	RLC charge	buoyancy
M	m	L	m
C	c	R	0 in the ideal model
K	k	$1/C$	$\rho g A$
u	x	q	x
F	$F(t)$	$E(t)$	0

Recognising the shared form allows the same characteristic-root and forcing methods to be reused without confusing the physical meanings.

The Complete Modeling Workflow

Step 1: Declare the unknown and positive direction. State what $u(t)$ measures and where zero is located.

Step 2: Convert all data to consistent units. Do not mix centimetres with metres, grams with kilograms, or weight with mass.

Step 3: List each force or voltage contribution. Attach a sign and physical law to every term before combining them.

Step 4: Construct the differential equation. Use Newton's law, Kirchhoff's law, or Archimedes' principle.

Step 5: Translate the initial data. Use the declared sign convention. In a circuit, remember $I = q'$.

Step 6: Classify before solving. Identify free or forced, damped or undamped, and the root regime when damping is present.

Step 7: Solve the complete IVP. Find $u_h + u_p$, calculate the required derivative, and apply all data.

Step 8: Interpret the result. Report units, transient and steady-state parts, frequency information, and long-term behavior.

Common Mistake: Confusing Mass With Weight

Newton's law uses mass:

$$mx'' = \sum F.$$

If a problem gives weight W, then:

$$W = mg, \qquad m = \frac{W}{g}.$$

Substituting weight directly for m gives the wrong inertial coefficient.

Common Mistake: Adding Gravity Twice

If x is measured from equilibrium, the balance $k\ell = mg$ has already removed gravity from the dynamic equation. The correct model is:

$$mx'' + cx' + kx = F(t).$$

Adding another mg term counts gravity twice.

Common Mistake: Losing The Velocity Sign

If downward is positive, an upward initial velocity is negative. If upward is positive, the same physical velocity is positive.

The phrase "moving upward" does not determine a numerical sign until the coordinate convention has been declared.

Common Mistake: Using The Wrong Damping Test

For:

$$mx'' + cx' + kx = 0,$$

the discriminant is:

$$c^2 - 4mk.$$

The coefficients must come from the unnormalized characteristic equation $mr^2 + cr + k = 0$. Omitting m changes the classification.

Common Mistake: Mixing Radians And Cycles

Circular frequency ω is measured in radians per second. Ordinary frequency is:

$$f = \frac{\omega}{2\pi}\ \text{Hz}.$$

The period is:

$$T = \frac{2\pi}{\omega}.$$

Writing $T = 1/\omega$ misses the factor 2π.

Common Mistake: Calling Every Forced Term Steady State

Steady state describes long-term behavior, not merely the fact that a term came from u_p.

For a stable damped system with bounded sinusoidal forcing, the particular sinusoidal response is steady state. In an undamped resonant system, a particular term such as $t\sin(\omega_0 t)$ grows and is not bounded steady-state motion.

Common Mistake: Using A One-Argument Phase Formula Blindly

The ratio B/A does not record the signs of A and B separately. A basic $\arctan(B/A)$ can therefore choose the wrong quadrant.

Use:

$$\cos\delta = \frac{A}{R}, \qquad \sin\delta = \frac{B}{R},$$

or a two-argument inverse tangent.

Block 7 Summary

The equation:

$$Mu'' + Cu' + Ku = F(t)$$

is useful only after its symbols have been tied to a physical model. Declare coordinates, convert units, derive signs, classify the system, solve the IVP, and interpret the result in the original units.

Original Practice Set

Practice Problems

Problem 1: Construct And Classify A Spring Model. A 5 kg mass moves vertically about its equilibrium position. The damping coefficient is 6 N s/m, the spring constant is 45 N/m, and an external force $20\cos(3t)$ N acts in the positive direction.

Take displacement in that direction as positive. At $t = 0$, the mass is 0.04 m from equilibrium and is moving in the opposite direction at 0.10 m/s.

Construct the initial-value problem. Then classify the motion as free or forced, damped or undamped, and underdamped, critically damped, or overdamped.

Problem 2: Find A Spring Constant From Static Stretch. A 4 kg mass stretches a vertical spring by 0.196 m. Use $g = 9.8\ \mathrm{m/s^2}$. After equilibrium has been established, the mass is pulled 0.05 m downward and released from rest. Ignore damping.

Find:

1. the spring constant;
2. the displacement from equilibrium;
3. the period of the motion.

Take downward displacement from equilibrium as positive.

Problem 3: Solve A Free Undamped IVP. A 2 kg mass is attached to a spring with spring constant 18 N/m. There is no damping or external force. With displacement measured positively downward, the initial data are:

$$x(0) = 0.10, \qquad x'(0) = -0.30.$$

Find the displacement $x(t)$ and its amplitude.

Problem 4: Read Harmonic Quantities From A Solution. For the motion:

$$x(t) = -0.12\cos(4t) + 0.05\sin(4t),$$

find:

1. the amplitude;
2. the circular frequency;
3. the ordinary frequency;
4. the period;
5. a phase angle δ such that $x = R\cos(4t - \delta)$.

Problem 5: Classify Three Damping Levels. For each of the following systems, classify the free motion as underdamped, critically damped, or overdamped:

$$m = 2, \qquad k = 8,$$

with:

$$\text{(a) } c = 4, \qquad \text{(b) } c = 8, \qquad \text{(c) } c = 10.$$

Problem 6: Solve A Free Damped IVP. Solve:

$$x'' + 6x' + 13x = 0,$$

subject to:

$$x(0) = 1, \qquad x'(0) = -3.$$

State the damping regime and identify the exponential envelope.

Problem 7: Find A Steady-State Response. Find a sinusoidal particular solution, and hence the steady-state response, for:

$$x'' + 4x' + 13x = 17\cos(2t) + \sin(2t).$$

You do not need to find the transient constants.

Problem 8: Solve A Resonant IVP. Solve:

$$x'' + 16x = 8\cos(4t),$$

subject to:

$$x(0) = 0, \qquad x'(0) = 0.$$

Explain why the usual trial $A\cos(4t) + B\sin(4t)$ must be modified.

Problem 9: Charge And Current In A Series RLC Circuit. A series circuit has:

$$L = 2\text{ H}, \qquad R = 4\ \Omega, \qquad C = \frac{1}{4}\text{ F},$$

and is driven by the constant voltage $E(t) = 8$ V. Initially:

$$q(0) = 1\text{ C}, \qquad I(0) = 0\text{ A}.$$

Find the charge $q(t)$ and the current $I(t)$.

Problem 10: Translate Between Mechanical And Electrical Models. An RLC charge equation is:

$$3q'' + 6q' + 12q = 9\cos t.$$

Write down a mechanically analogous spring equation by identifying the corresponding mass, damping coefficient, spring constant, and external force. Classify the homogeneous response of both systems.

Problem 11: Model A Floating Cylinder. A vertical cylinder of mass 2 kg and horizontal cross-sectional area 0.01 m^2 floats in water of mass density 1000 $\mathrm{kg/m}^3$. Use $g = 9.8\ \mathrm{m/s}^2$ and ignore resistance.

Take upward displacement from equilibrium as positive. The cylinder is pushed 0.03 m downward and released from rest.

Find:

1. the equilibrium submerged depth;
2. the differential equation for displacement;
3. the resulting motion;
4. the period.

Problem 12: Diagnose Four Modelling Errors. Correct each statement.

1. An object weighing 98 N has mass 98 kg.
2. If the circular frequency is 5 rad/s, the period is 1/5 s.
3. If downward is positive and a mass initially moves upward at 0.20 m/s, then $x'(0) = 0.20$.
4. For $x'' + 9x = \cos(3t)$, a suitable particular trial is $A\cos(3t) + B\sin(3t)$.

Worked Answers: Problems 1 To 4

Answer 1. The mass-spring model measured from equilibrium is:

$$mx'' + cx' + kx = F(t).$$

Insert the given coefficients:

$$5x'' + 6x' + 45x = 20\cos(3t).$$

The initial displacement is positive:

$$x(0) = 0.04.$$

The mass initially moves opposite to the declared positive direction, so its initial velocity is negative:

$$x'(0) = -0.10.$$

Therefore the complete initial-value problem is:

$$\boxed{5x'' + 6x' + 45x = 20\cos(3t), \qquad x(0) = 0.04, \qquad x'(0) = -0.10}.$$

The right side is not zero, so the motion is **forced**. Since $c = 6 > 0$, it is also **damped**.

To classify the homogeneous response, use the discriminant from:

$$5r^2 + 6r + 45 = 0.$$

Thus:

$$\begin{aligned}\Delta &= c^2 - 4mk \\ &= 6^2 - 4(5)(45) \\ &= 36 - 900 \\ &= -864.\end{aligned}$$

Since $\Delta < 0$, the homogeneous response is **underdamped**. The complete classification is forced, damped motion with an underdamped transient.

Answer 2. At static equilibrium, the spring force balances the weight:

$$k\ell = mg.$$

Here:

$$m = 4, \qquad \ell = 0.196, \qquad g = 9.8.$$

Substitute these values:

$$k(0.196) = 4(9.8).$$

The right side is:

$$4(9.8) = 39.2.$$

Divide by 0.196:

$$\boxed{k = 200\ \text{N/m}}.$$

Displacement is measured from equilibrium, so gravity has already been cancelled by the equilibrium spring stretch. With no damping or forcing:

$$mx'' + kx = 0.$$

Insert $m = 4$ and $k = 200$:

$$4x'' + 200x = 0.$$

Divide by 4:

$$x'' + 50x = 0.$$

The circular frequency is:

$$\omega_0 = \sqrt{50} = 5\sqrt{2}.$$

Therefore:

$$x = c_1 \cos(5\sqrt{2}\,t) + c_2 \sin(5\sqrt{2}\,t).$$

The mass is pulled 0.05 m downward, and downward is positive:

$$x(0) = 0.05.$$

Substitute $t = 0$:

$$0.05 = c_1.$$

Differentiate the solution:

$$x' = -5\sqrt{2}\,c_1 \sin(5\sqrt{2}\,t) + 5\sqrt{2}\,c_2 \cos(5\sqrt{2}\,t).$$

The mass is released from rest, so:

$$x'(0) = 0.$$

Substitute $t = 0$:

$$0 = 5\sqrt{2}\,c_2.$$

Hence:

$$c_2 = 0.$$

The displacement is:

$$\boxed{x(t) = 0.05\cos(5\sqrt{2}\,t)\ \mathrm{m}}.$$

Finally:

$$T = \frac{2\pi}{\omega_0}.$$

Substitute $\omega_0 = 5\sqrt{2}$:

$$\boxed{T = \frac{2\pi}{5\sqrt{2}}\ \mathrm{s}}.$$

Answer 3. With no damping or external force, the model is:

$$mx'' + kx = 0.$$

Insert $m = 2$ and $k = 18$:

$$2x'' + 18x = 0.$$

Divide by 2:

$$x'' + 9x = 0.$$

The characteristic equation is:

$$r^2 + 9 = 0,$$

so the roots are:

$$r = \pm 3i.$$

Therefore:

$$x = c_1 \cos(3t) + c_2 \sin(3t).$$

Apply the initial displacement:

$$\begin{aligned} 0.10 &= c_1 \cos(0) + c_2 \sin(0) \\ &= c_1. \end{aligned}$$

Thus:

$$c_1 = 0.10.$$

Differentiate:

$$x' = -3c_1 \sin(3t) + 3c_2 \cos(3t).$$

Apply the initial velocity:

$$\begin{aligned} -0.30 &= -3c_1 \sin(0) + 3c_2 \cos(0) \\ &= 3c_2. \end{aligned}$$

Divide by 3:

$$c_2 = -0.10.$$

The displacement is:

$$\boxed{x(t) = 0.10 \cos(3t) - 0.10 \sin(3t) \text{ m}}.$$

For $A \cos(\omega t) + B \sin(\omega t)$, the amplitude is:

$$R = \sqrt{A^2 + B^2}.$$

Here $A = 0.10$ and $B = -0.10$, so:

$$\begin{aligned} R &= \sqrt{(0.10)^2 + (-0.10)^2} \\ &= \sqrt{0.02} \\ &= 0.10\sqrt{2}. \end{aligned}$$

Therefore:

$$\boxed{R = 0.10\sqrt{2}\,\mathrm{m}}.$$

Answer 4. Compare:

$$x(t) = -0.12\cos(4t) + 0.05\sin(4t)$$

with:

$$x(t) = A\cos(\omega t) + B\sin(\omega t).$$

Thus:

$$A = -0.12, \qquad B = 0.05, \qquad \omega = 4.$$

The amplitude is:

$$\begin{aligned} R &= \sqrt{A^2 + B^2} \\ &= \sqrt{(-0.12)^2 + (0.05)^2} \\ &= \sqrt{0.0144 + 0.0025} \\ &= \sqrt{0.0169} \\ &= 0.13. \end{aligned}$$

Therefore:

$$\boxed{R = 0.13}.$$

The circular frequency is:

$$\boxed{\omega = 4 \text{ rad/s}}.$$

Convert circular frequency to ordinary frequency:

$$f = \frac{\omega}{2\pi}.$$

Substitute $\omega = 4$:

$$\boxed{f = \frac{2}{\pi} \text{ Hz}}.$$

The period is:

$$T = \frac{2\pi}{\omega}.$$

Therefore:

$$\boxed{T = \frac{\pi}{2} \text{ s}}.$$

To write $x = R\cos(4t - \delta)$, expand:

$$R\cos(4t - \delta) = R\cos\delta\cos(4t) + R\sin\delta\sin(4t).$$

Match coefficients with the given formula:

$$R\cos\delta = -0.12, \qquad R\sin\delta = 0.05.$$

Since $R = 0.13$:

$$\cos\delta = -\frac{12}{13}, \qquad \sin\delta = \frac{5}{13}.$$

Cosine is negative and sine is positive, so δ lies in quadrant II. One suitable phase angle is:

$$\boxed{\delta = \pi - \arctan\left(\frac{5}{12}\right)}.$$

Worked Answers: Problems 5 To 8

Answer 5. For free damped motion:

$$mx'' + cx' + kx = 0,$$

the characteristic discriminant is:

$$\Delta = c^2 - 4mk.$$

Here:

$$m = 2, \qquad k = 8.$$

Therefore:

$$\begin{aligned} \Delta &= c^2 - 4(2)(8) \\ &= c^2 - 64. \end{aligned}$$

For $c = 4$:

$$\begin{aligned} \Delta &= 4^2 - 64 \\ &= 16 - 64 \\ &= -48. \end{aligned}$$

Since $\Delta < 0$, case (a) is:

$$\boxed{\text{underdamped}}.$$

For $c = 8$:

$$\begin{aligned}\Delta &= 8^2 - 64\\ &= 64 - 64\\ &= 0.\end{aligned}$$

Since $\Delta = 0$, case (b) is:

$$\boxed{\text{critically damped}}.$$

For $c = 10$:

$$\begin{aligned}\Delta &= 10^2 - 64\\ &= 100 - 64\\ &= 36.\end{aligned}$$

Since $\Delta > 0$, case (c) is:

$$\boxed{\text{overdamped}}.$$

Answer 6. The characteristic equation for:

$$x'' + 6x' + 13x = 0$$

is:

$$r^2 + 6r + 13 = 0.$$

Use the quadratic formula:

$$r = \frac{-6 \pm \sqrt{6^2 - 4(1)(13)}}{2}.$$

Simplify the discriminant:

$$6^2 - 4(13) = 36 - 52 = -16.$$

Therefore:

$$\begin{aligned} r &= \frac{-6 \pm \sqrt{-16}}{2} \\ &= \frac{-6 \pm 4i}{2} \\ &= -3 \pm 2i. \end{aligned}$$

The roots are complex with a negative real part, so the system is **under-damped**. Its general solution is:

$$x = e^{-3t}(c_1 \cos(2t) + c_2 \sin(2t)).$$

Apply $x(0) = 1$:

$$\begin{aligned} 1 &= e^0(c_1 \cos(0) + c_2 \sin(0)) \\ &= c_1. \end{aligned}$$

Thus:

$$c_1 = 1.$$

Differentiate the product:

$$\begin{aligned} x' = & -3e^{-3t}(c_1 \cos(2t) + c_2 \sin(2t)) \\ & + e^{-3t}(-2c_1 \sin(2t) + 2c_2 \cos(2t)). \end{aligned}$$

At $t = 0$:

$$x'(0) = -3c_1 + 2c_2.$$

Insert $x'(0) = -3$ and $c_1 = 1$:

$$-3 = -3(1) + 2c_2.$$

Add 3 to both sides:

$$0 = 2c_2.$$

Therefore:

$$c_2 = 0.$$

The solution is:

$$\boxed{x(t) = e^{-3t} \cos(2t)}.$$

Because $-1 \le \cos(2t) \le 1$, the oscillation lies between:

$$-e^{-3t} \qquad \text{and} \qquad e^{-3t}.$$

Hence the exponential envelopes are:

$$\boxed{x = \pm e^{-3t}}.$$

Answer 7. Seek a particular solution with the same two sinusoidal building blocks as the forcing:

$$x_p = A\cos(2t) + B\sin(2t).$$

Differentiate once:

$$x_p' = -2A\sin(2t) + 2B\cos(2t).$$

Differentiate again:

$$x_p'' = -4A\cos(2t) - 4B\sin(2t).$$

Substitute all three expressions into:

$$x'' + 4x' + 13x = 17\cos(2t) + \sin(2t).$$

This gives:

$$\begin{aligned}
&[-4A\cos(2t) - 4B\sin(2t)] \\
&\quad + 4[-2A\sin(2t) + 2B\cos(2t)] \\
&\quad + 13[A\cos(2t) + B\sin(2t)] \\
&= 17\cos(2t) + \sin(2t).
\end{aligned}$$

Collect the cosine terms:

$$-4A + 8B + 13A = 9A + 8B.$$

Collect the sine terms:

$$-4B - 8A + 13B = -8A + 9B.$$

Therefore the substituted equation is:

$$(9A + 8B)\cos(2t) + (-8A + 9B)\sin(2t) = 17\cos(2t) + \sin(2t).$$

Match the cosine and sine coefficients:

$$\begin{aligned} 9A + 8B &= 17, \\ -8A + 9B &= 1. \end{aligned}$$

Multiply the first equation by 9:

$$81A + 72B = 153.$$

Multiply the second equation by 8:

$$-64A + 72B = 8.$$

Subtract the second equation from the first:

$$145A = 145.$$

Thus:

$$A = 1.$$

Substitute $A = 1$ into $9A + 8B = 17$:

$$9 + 8B = 17.$$

Subtract 9:

$$8B = 8.$$

Therefore:

$$B = 1.$$

The sinusoidal particular solution is:

$$x_p = \cos(2t) + \sin(2t).$$

Since the homogeneous roots have negative real part, the homogeneous transient decays. The steady-state response is therefore:

$$\boxed{x_{\text{ss}}(t) = \cos(2t) + \sin(2t)}.$$

Answer 8. The homogeneous equation:

$$x'' + 16x = 0$$

has the solution:

$$x_h = c_1 \cos(4t) + c_2 \sin(4t).$$

The forcing $\cos(4t)$ already appears in the homogeneous solution. Therefore the usual trial:

$$A \cos(4t) + B \sin(4t)$$

would be absorbed into x_h and could not produce the nonzero right side. This is resonance, so multiply the trial by t.

For this forcing, use the simpler resonant trial:

$$x_p = Ct \sin(4t).$$

Differentiate once:

$$x_p' = C \sin(4t) + 4Ct \cos(4t).$$

Differentiate again:

$$x_p'' = 4C \cos(4t) + 4C \cos(4t) - 16Ct \sin(4t).$$

Combine the two cosine terms:

$$x_p'' = 8C\cos(4t) - 16Ct\sin(4t).$$

Now substitute into the left side:

$$\begin{aligned} x_p'' + 16x_p &= 8C\cos(4t) - 16Ct\sin(4t) + 16Ct\sin(4t) \\ &= 8C\cos(4t). \end{aligned}$$

Match this with the forcing $8\cos(4t)$:

$$8C\cos(4t) = 8\cos(4t).$$

Thus:

$$C = 1.$$

The complete solution is:

$$x = c_1\cos(4t) + c_2\sin(4t) + t\sin(4t).$$

Apply $x(0) = 0$:

$$0 = c_1.$$

Differentiate the complete solution:

$$x' = -4c_1\sin(4t) + 4c_2\cos(4t) + \sin(4t) + 4t\cos(4t).$$

Apply $x'(0) = 0$:

$$0 = 4c_2.$$

Therefore:

$$c_2 = 0.$$

The resonant motion is:

$$\boxed{x(t) = t\sin(4t)}.$$

The factor t makes the oscillation amplitude grow with time.

Worked Answers: Problems 9 To 12

Answer 9. The charge equation for a series RLC circuit is:

$$Lq'' + Rq' + \frac{1}{C}q = E(t).$$

Insert:

$$L = 2, \qquad R = 4, \qquad C = \frac{1}{4}, \qquad E(t) = 8.$$

Since:

$$\frac{1}{C} = \frac{1}{1/4} = 4,$$

the equation becomes:

$$2q'' + 4q' + 4q = 8.$$

Divide every term by 2:

$$q'' + 2q' + 2q = 4.$$

The homogeneous characteristic equation is:

$$r^2 + 2r + 2 = 0.$$

Use the quadratic formula:

$$\begin{aligned} r &= \frac{-2 \pm \sqrt{2^2 - 4(1)(2)}}{2} \\ &= \frac{-2 \pm \sqrt{-4}}{2} \\ &= -1 \pm i. \end{aligned}$$

Therefore:

$$q_h = e^{-t}(c_1 \cos t + c_2 \sin t).$$

Because the applied voltage is constant, try a constant particular charge:

$$q_p = Q.$$

Then:

$$q_p' = 0, \qquad q_p'' = 0.$$

Substitute into $q'' + 2q' + 2q = 4$:

$$2Q = 4.$$

Divide by 2:

$$Q = 2.$$

Thus the complete charge is:

$$q = 2 + e^{-t}(c_1 \cos t + c_2 \sin t).$$

Apply $q(0) = 1$:

$$\begin{aligned} 1 &= 2 + e^{0}(c_1 \cos 0 + c_2 \sin 0) \\ &= 2 + c_1. \end{aligned}$$

Subtract 2:

$$c_1 = -1.$$

Current is the derivative of charge:

$$I = q'.$$

Differentiate the complete charge:

$$\begin{aligned} I = & -e^{-t}(c_1 \cos t + c_2 \sin t) \\ & + e^{-t}(-c_1 \sin t + c_2 \cos t). \end{aligned}$$

At $t = 0$:

$$I(0) = -c_1 + c_2.$$

Use $I(0) = 0$ and $c_1 = -1$:

$$0 = -(-1) + c_2.$$

Therefore:

$$c_2 = -1.$$

The charge is:

$$\boxed{q(t) = 2 - e^{-t}(\cos t + \sin t)\ \mathrm{C}}.$$

Differentiate this final formula:

$$\begin{aligned} I(t) &= -\frac{d}{dt}\left[e^{-t}(\cos t + \sin t)\right] \\ &= -e^{-t}\left[-(\cos t + \sin t) - \sin t + \cos t\right] \\ &= -e^{-t}(-2\sin t). \end{aligned}$$

Hence:

$$\boxed{I(t) = 2e^{-t}\sin t \text{ A}}.$$

As $t \to \infty$, the exponential terms vanish, so $q \to 2$ C and $I \to 0$ A.

Answer 10. The series RLC charge model is:

$$Lq'' + Rq' + \frac{1}{C}q = E(t).$$

The mechanical spring model is:

$$mx'' + cx' + kx = F(t).$$

Match terms in:

$$3q'' + 6q' + 12q = 9\cos t.$$

The corresponding quantities are:

$$m \longleftrightarrow L = 3,$$

$$c \longleftrightarrow R = 6,$$

$$k \longleftrightarrow \frac{1}{C} = 12,$$

and:

$$F(t) \longleftrightarrow E(t) = 9\cos t.$$

Therefore a mechanically analogous equation is:

$$\boxed{3x'' + 6x' + 12x = 9\cos t}.$$

For either homogeneous system, the characteristic equation is:

$$3r^2 + 6r + 12 = 0.$$

Its discriminant is:

$$\begin{aligned}\Delta &= 6^2 - 4(3)(12)\\ &= 36 - 144\\ &= -108.\end{aligned}$$

Since $\Delta < 0$, the mechanical system has an underdamped transient. By the same coefficient structure, the circuit also has an oscillatory transient whose amplitude decays exponentially.

Answer 11. At equilibrium, the buoyant force equals the weight. If h is the equilibrium submerged depth, then:

$$\rho g A h = m g.$$

Cancel g from both sides:

$$\rho A h = m.$$

Solve for h:

$$h = \frac{m}{\rho A}.$$

Insert:

$$m = 2, \qquad \rho = 1000, \qquad A = 0.01.$$

Therefore:

$$\begin{aligned} h &= \frac{2}{1000(0.01)} \\ &= \frac{2}{10} \\ &= 0.20. \end{aligned}$$

Thus:

$$\boxed{h = 0.20 \text{ m}}.$$

Now let x be upward displacement from equilibrium. An upward displacement x reduces the submerged depth from h to:

$$h - x.$$

The upward buoyant force is then:

$$\rho g A(h - x).$$

Weight acts downward, so Newton's second law is:

$$mx'' = \rho g A(h - x) - mg.$$

Expand the buoyancy term:

$$mx'' = \rho g A h - \rho g A x - mg.$$

At equilibrium:

$$\rho g A h = mg.$$

Substitute that equality into the force equation:

$$mx'' = mg - \rho g A x - mg.$$

The two constant weight terms cancel:

$$mx'' = -\rho g A x.$$

Move the restoring term to the left:

$$mx'' + \rho g A x = 0.$$

Insert the numerical values:

$$2x'' + 1000(9.8)(0.01)x = 0.$$

Calculate the coefficient of x:

$$1000(9.8)(0.01) = 98.$$

Therefore:

$$2x'' + 98x = 0.$$

Divide by 2:

$$\boxed{x'' + 49x = 0}.$$

The circular frequency is:

$$\omega_0 = \sqrt{49} = 7.$$

Hence:

$$x = c_1 \cos(7t) + c_2 \sin(7t).$$

Upward is positive, but the cylinder is initially pushed 0.03 m downward. Therefore:

$$x(0) = -0.03.$$

Substitute $t = 0$:

$$c_1 = -0.03.$$

Differentiate:

$$x' = -7c_1 \sin(7t) + 7c_2 \cos(7t).$$

The cylinder is released from rest:

$$x'(0) = 0.$$

Thus:

$$0 = 7c_2,$$

so:

$$c_2 = 0.$$

The motion is:

$$\boxed{x(t) = -0.03\cos(7t)\ \text{m}}.$$

Finally:

$$T = \frac{2\pi}{\omega_0}.$$

Substitute $\omega_0 = 7$:

$$\boxed{T = \frac{2\pi}{7}\ \mathrm{s}}.$$

Answer 12.

Statement 1. Weight and mass are not the same quantity. They are related by:

$$W = mg.$$

For $W = 98$ N and $g = 9.8\ \mathrm{m/s^2}$:

$$m = \frac{W}{g}.$$

Substitute the values:

$$m = \frac{98}{9.8} = 10.$$

The corrected statement is:

$$\boxed{\text{An object weighing 98 N has mass 10 kg.}}$$

Statement 2. Circular frequency and period satisfy:

$$T = \frac{2\pi}{\omega}.$$

For $\omega = 5$ rad/s:

$$\boxed{T = \frac{2\pi}{5}\ \mathrm{s}}.$$

The proposed $1/5$ omits the factor 2π.

Statement 3. The coordinate declares downward velocity positive. An upward velocity must therefore be negative.

The corrected initial condition is:

$$\boxed{x'(0) = -0.20\ \mathrm{m/s}}.$$

Statement 4. The homogeneous equation:

$$x'' + 9x = 0$$

has the basis:

$$\cos(3t), \qquad \sin(3t).$$

Therefore the trial $A\cos(3t) + B\sin(3t)$ duplicates the homogeneous solution. The forcing frequency equals the natural frequency, so this is resonance.

Multiply the usual sinusoidal trial by t. One suitable choice is:

$$\boxed{x_p = Ct\sin(3t)}.$$

Chapter 14 Final Summary

A physical second-order model begins with a stated unknown, a coordinate direction, and a balance law. For a mass-spring system measured from equilibrium:

$$\boxed{mx'' + cx' + kx = F(t)}.$$

Its free response is classified using:

$$\boxed{\Delta = c^2 - 4mk}.$$

The three possibilities are:

$$\begin{aligned}
\Delta > 0 &\quad \Longrightarrow \text{overdamped,}\\
\Delta = 0 &\quad \Longrightarrow \text{critically damped,}\\
\Delta < 0 &\quad \Longrightarrow \text{underdamped.}
\end{aligned}$$

Without damping, the natural circular frequency and period are:

$$\boxed{\omega_0 = \sqrt{\frac{k}{m}}, \qquad T = \frac{2\pi}{\omega_0}}.$$

For stable forced damped motion, the homogeneous part is the transient and the bounded particular response is the steady state. If undamped forcing matches the natural frequency, resonance introduces a growing factor of t.

A series RLC circuit follows:

$$\boxed{Lq'' + Rq' + \frac{1}{C}q = E(t), \qquad I = q'}.$$

A floating vertical cylinder with no resistance follows:

$$\boxed{mx'' + \rho g A x = 0}.$$

These models share the structure:

$$\boxed{Mu'' + Cu' + Ku = F(t)}.$$

The transferable method is:

1. define the unknown and positive direction;
2. convert every quantity to compatible units;
3. construct the balance law term by term;
4. translate the initial data with the declared sign convention;
5. classify the homogeneous response;
6. solve the complete initial-value problem;
7. interpret the formula in the original physical setting.

The central lesson is:

the same differential-equation structure can describe very different systems, but the symbols gain meaning only after the physical law, signs, units, and initial data have been made explicit.